

Disclaimer

- This training resource was developed to be used with the CZ ID platform
- In October, 2026, stewardship of CZ ID transitioned to UCSF and the platform was renamed to SeqtoID
- These training materials are not being actively maintained and may not reflect differences between CZ ID and SeqtoID, subsequent changes to the platform, or current methods and best practices in metagenomic sequencing and analysis



Sample Report Metrics

Sample Report

< ASTMH_Demos

Nanopore mNGS Pipeline v0.7 | NCBI Index Date: 2021-01-22 | processed a year ago | [SAMPLE DETAILS](#)

SF_Mission_barcode11_392772_original ▾

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Nanopore

Taxon name

🔍

Name Type: Scientific ▾

Categories ▾

Threshold filters ▾

Read Specificity: Specific Only ▾

Annotation ▾

Pathogen Flags ▾

241 rows passing the above filters, out of 259 total rows

📄

🔍

>	Taxon	bPM ▾	b	r	contig	cont...	%id	L	E val...	NT NR
> ▢	Alphacoronavirus (1 viral species: ● 1)	298,841.0 280,938.0	28,011,475 26,333,312	59,591 57,854	13 13	26,333,312 26,333,312	98.7 97.5	1,896.0 597.9	10 ⁻³⁰⁶ 10 ⁻³⁰³	
> ▢	Corynebacterium (34 bacterial species: ● 4)	25,957.4 17,674.9	2,433,077 1,656,731	2,130 1,242	2 2	1,656,731 1,656,731	95.8 76.6	1,597.6 234.3	10 ⁻²⁷³ 10 ⁻¹⁰⁴	
> ▢	Malassezia (2 eukaryotic species)	4,794.7 3,333.7	449,422 312,481	1,891 1,756	1 1	312,481 312,481	97.7 97.4	1,743.9 582.0	10 ⁻³⁰⁵ 10 ⁻³⁰⁸	
> ▢	Klebsiella (4 bacterial species: ● 2)	3,045.6 0.0	285,479 0	194 0	1 0	272,838 0	98.7 0.0	2,560.2 0.0	10 ⁻²⁹⁹ 0	

Sample Report Metrics

Bases per million (bPM): Number of bases within all the reads aligning to a given taxon, including those assembled into contigs that mapped to the taxon, per million bases sequenced

Bases (b): Number of bases within all the reads aligning to a given taxon, including those assembled into contigs that mapped to the taxon.

Reads (r): Number of reads aligning to a given taxon, including those assembled into contigs that mapped to the taxon.

Contig: Number of assembled contigs aligning to a given taxon.

Contig bases (contig b): Number of bases within all the reads that assembled into contigs aligning to a given taxon.

Percent identity (% ID): Average percent identity between all the query sequences (contigs and unassembled reads) and their matching taxon

Length (L): Average length of alignments between all the query sequences (contigs and unassembled reads) and their matching taxon (values against NR are reported in base pairs)

Sample Report: each metric shows two results (NT/NR)

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Taxon name Name Type: Scientific ▾ Categories ▾ Threshold filters ▾ Read Specificity: Specific Only ▾ Annotation ▾ Pathogen Flags ▾

241 rows passing the above filters, out of 259 total rows

NT = nucleotide database
NR = protein database

	Taxon	bPM ▾	b	r	contig	cont...	%id	L	E val...	NT NR
>	Alphacoronavirus (1 viral species: ● 1)	298,841.0 280,938.0	28,011,475 26,333,312	59,591 57,854	13 13	26,333,312 26,333,312	98.7 97.5	1,896.0 597.9	10 ⁻³⁰⁶ 10 ⁻³⁰³	
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>	Klebsiella (4 bacterial species: ● 2)	3,045.6 0.0	285,479 0	194 0	1 0	272,838 0	98.7 0.0	2,560.2 0.0	10 ⁻²⁹⁹ 0	

Important: Values against NR only reflect alignments between *contig* sequences and their matching taxon (i.e., *unassembled reads are not aligned against NR*).

General Interpretation Notes:

- Contigs aligning to both databases: Good sign that the hit is true (more to it than that)
- If a contig is viral and only has a hit to the NR database: Hit could represent a divergent virus
- If a hit is bacterial and it only has a contig to the NT database: Sequences may be aligning to the 16S rRNA region, which is not in the NR databases (should not be trusted)

Sample Report

< Nanopore_Test

SRR22762466_2

Nanopore mNGS Pipeline v0.6 | processed 6 months ago | [SAMPLE DETAILS](#)

Metagenomics - Nanopore

BETA

Taxon name

Name Type: Scientific

Categories

Threshold filters

Read Specificity: All

Annotation

736 rows

	Taxon	bPM	b	r	contig	cont...	%id	L	E val...	NT NR
>	Leuconostoc (10 bacterial species:)	29,702.0 457.2	2,743,794 42,236	1,586 17	2 2	432,151 42,236	90.3 84.2	1,650.3 804.1	10 ⁻³⁰⁰ 10 ⁻²⁸⁰	
>	Lactiplantibacillus (3 bacterial species:)	24,661.3 232.5	2,278,146 21,482	1,091 11	2 2	113,076 21,482	90.5 89.8	2,134.2 125.7	10 ⁻³⁰⁵ 10 ⁻⁵⁹	
>	Chryseobacterium (6 bacterial species:)	13,031.3 160.7	1,203,802 14,845	241 3	1 1	14,845 14,845	88.7 74.5	4,622.4 564.0	10 ⁻³⁰⁷ 10 ⁻²³⁹	
>	Citrobacter (11 bacterial species: ● 1)	12,972.7 0.0	1,198,382 0	186 0	0 0	0 0	89.4 0.0	6,212.9 0.0	10 ⁻³⁰⁷ 0	
>	Pseudomonas (22 bacterial species: ● 3)	9,651.9 0.0	891,617 0	144 0	0 0	0 0	86.8 0.0	4,443.7 0.0	10 ⁻³⁰⁶ 0	

Important: Values against NR only reflect alignments between *contig* sequences and their matching taxon (i.e., *unassembled reads are not aligned against NR*).

No contigs = no chance for NR hit

Filters are helpful at removing false hits, but not perfect

ASTMH_Demos Nanopore mNGS Pipeline v0.7 | NCBI Index Date: 2021-01-22 | processed a year ago | [SAMPLE DETAILS](#)

SF_Mission_barcode11_392772_original ▾

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Nanopore

Taxon name Name Type: Scientific ▾ Categories ▾ Threshold filters ▾ Read Specificity: Specific Only ▾ Annotation ▾ Pathogen Flags ▾

241 rows passing the above filters, out of 259 total rows

>	Taxon	bPM ▾	b	r	contig	cont...	%id	L	E val...	NT NR
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▾	Human coronavirus 229E Known Pathogen 📊 ⬇	298,841.0 280,938.0	28,011,475 26,333,312	59,591 57,854	13 13	26,333,312 26,333,312	98.7 97.5	1,896.0 597.9	10 ⁻³⁰⁶ 10 ⁻³⁰³	
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Suggested (initial) filters:

- bPM NT > 100 to remove taxa that were present at low levels.
- bPM NR > 1 to remove taxa that only have matches in non-coding regions **(this will only help if there are contigs)**
- NT L > 75 bp to remove taxa with alignments < 75 bp long
- E-value (NT and NR) < 0.001 to remove likely random matches. In general, we suggest setting the E-value filter to $\leq 10^{-10}$ for stringent searches or ≤ 0.01 to allow for less significant taxon matches.

Gaining confidence in taxon calls using report metrics

Metric	Definition	Interpretation
bPM	Number of bases within the reads aligning to taxon per million bases sequenced (normalized metric)	Higher bPM = more confidence in the hit
Bases (b)	Number of bases within the reads aligning to taxon	Higher b = more confidence in the hit
Reads (r)	Number of reads aligning to taxon	Higher r = more confidence in the hit
Contig	Number of assembled contigs aligning to taxon	Longer contig = more confidence
Contig r	Number of reads aligning to assembled contigs	Higher Contig r = more confidence
% id	Average % identity of alignment to accession	Should be taken into account with other metrics
L	Average length of alignment of contigs and reads	Longer alignment = more confidence (if it's divergent, L may not be high)
E value	Average expect value of alignment	Lower = more confidence

Available Analysis Icons

< ASTMH_Demos Nanopore mNGS Pipeline v0.7 | NCBI Index Date: 2021-01-22 | processed a year ago | [SAMPLE DETAILS](#)

SF_Mission_barcode11_392772_original ▾

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Nanopore

Taxon name Name Type: Scientific ▾ Categories ▾ Threshold filters ▾ Read Specificity: Specific Only ▾ Annotation ▾ Pathogen Flags ▾

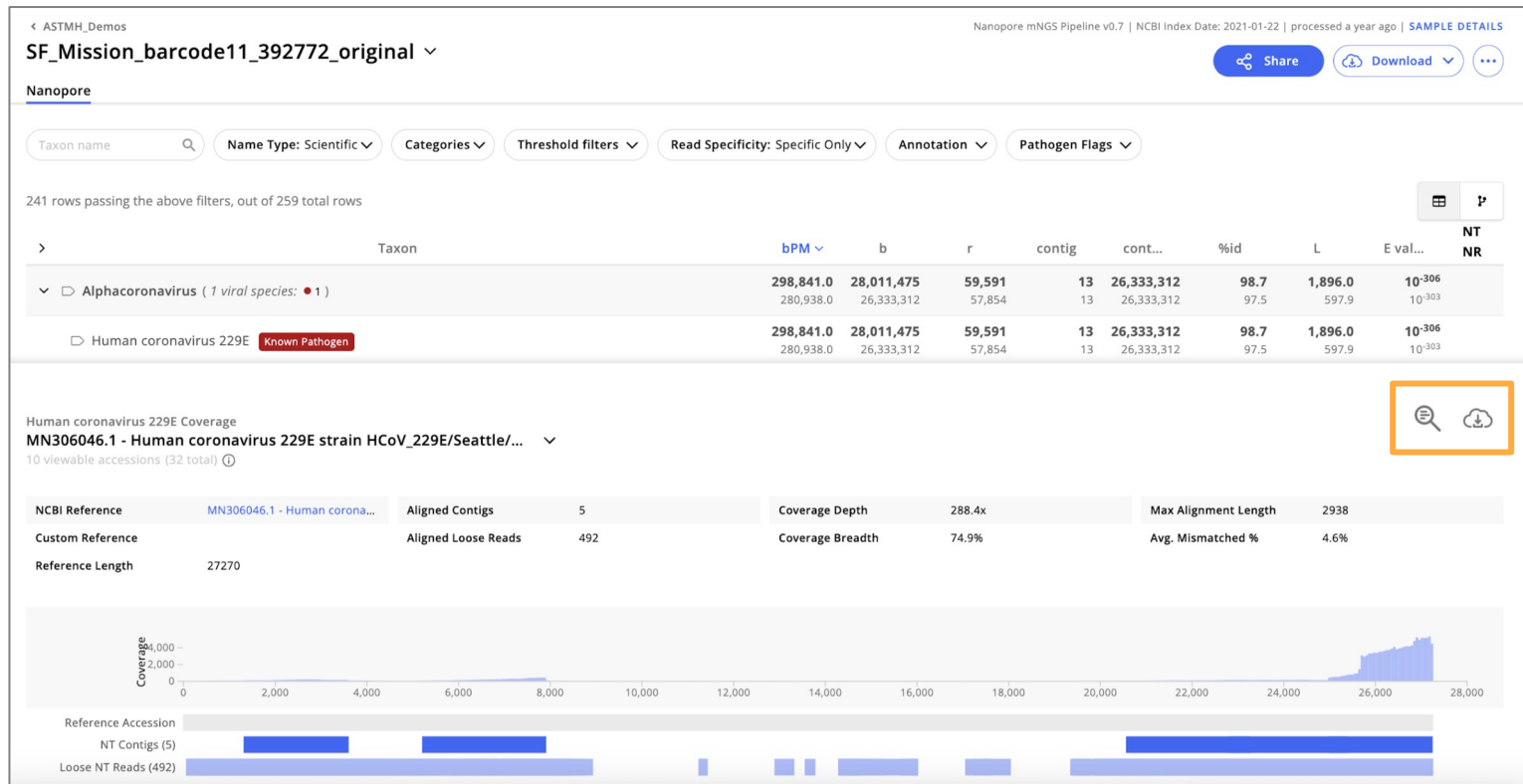
241 rows passing the above filters, out of 259 total rows

>	Taxon	bPM ▾	b	r	contig	cont...	%id	L	E val...	NT NR
▾	Alphacoronavirus (1 viral species: ● 1)	298,841.0 280,938.0	28,011,475 26,333,312	59,591 57,854	13 13	26,333,312 26,333,312	98.7 97.5	1,896.0 597.9	10 ⁻³⁰⁶ 10 ⁻³⁰³	
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 **Coverage visualization:** Available for any taxon supported by > 1 read matching NT

 **Download:** Option to download read or contig sequences (FASTA) associated with a given taxon

Available Analysis Icons (coverage visualization panel)



-  **BLAST:** Only available for taxa with at least one contig matching NT.
-  **Download:** Download contig sequences (FASTA).

The coverage visualization is valuable for evaluating hits

Alphapapillomavirus 7 Coverage

MF288662.1 - Human papillomavirus type 18 isolate 1359486_N-P, c... ▾

10 viewable accessions (46 total) ⓘ

[View read-level visualization >](#)



Reference NCBI Entry [MF288662.1 - Human p...](#)

Aligned Contigs 1

Coverage Depth 936.8x

Max Alignment Length 7857

Reference Length 7857

Aligned Loose Reads 0

Coverage Breadth 100.0%

Avg. Mismatched % 0.0%

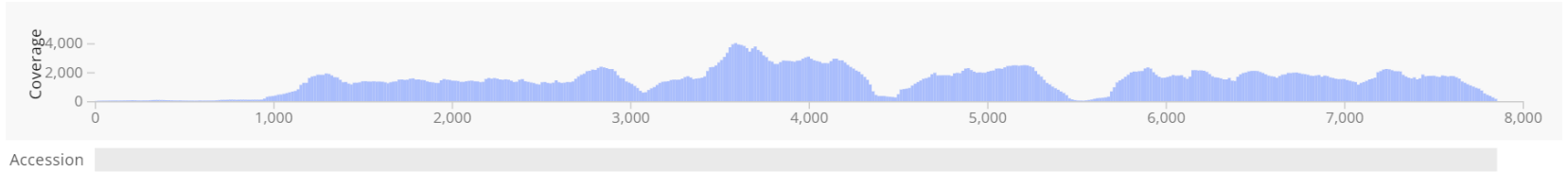


Reference Accession

Contigs (1)

Take into consideration the depth and breadth of the coverage (the higher, the better)

depth



breadth

Taking into account coverage breadth and depth

Depth	Breadth	How to Interpret
High	High	Provides you with confidence that this is a true hit!
Low	High	If there is coverage across multiple regions of the genome, this is likely a real hit. Evaluate the assembled contigs (NT L) and ensure the alignment is good (% identity).
High	Low	Depends on whether the small portion that has high coverage is unique to that particular taxon and on the quality of the alignment and assembly.
Low	Low	May be a result of genomic similarity at the 16S rRNA region (for bacteria)

What do we think about this viral hit?

Chikungunya virus Coverage

MK468610.1 - Chikungunya virus isolate CHRF_0094_11-22-2017, co... ▾

10 viewable accessions (79 total) ⓘ



NCBI Reference [MK468610.1 - Chikungunya vi...](#)

Aligned Contigs 2

Coverage Depth 3415.7x

Max Alignment Length 11818

Custom Reference

Aligned Loose Reads 99

Coverage Breadth 100.0%

Avg. Mismatched % 0.6%

Reference Length 11818



Reference Accession

NT Contigs (2)

Loose NT Reads (99)



What do we think about this bacterial hit?

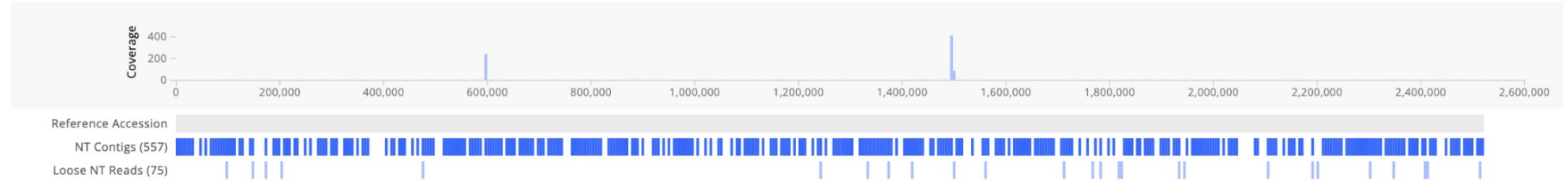
Cutibacterium acnes Coverage

CP003196.1 - Propionibacterium acnes Type1A2 P.acn17, complete ...

16 viewable accessions (32 total)



NCBI Reference	CP003196.1 - Propionibacteri...	Aligned Contigs	557	Coverage Depth	1.7x	Max Alignment Length	3227
Custom Reference		Aligned Loose Reads	75	Coverage Breadth	7.0%	Avg. Mismatched %	0.3%
Reference Length	2522885						



What do we think about this bacterial hit?

Providencia rettgeri Coverage

[View read-level visualization >](#)

CP029736.1 - *Providencia rettgeri* strain AR_0082 chromosome, co... ▼

1 viewable accessions



Reference NCBI Entry [CP029736.1 - Providenc...](#)

Aligned Contigs

1

Coverage Depth

0.4x

Max Alignment Length 1354

Reference Length 4309166

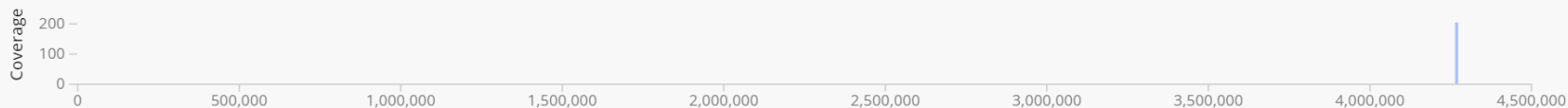
Aligned Loose Reads

0

Coverage Breadth

0.0%

Avg. Mismatched % 0.2%



Reference Accession

Contigs (1)

BLAST can be used for hit confirmation

BLASTN Contig

Escherichia

BLASTN compares nucleotide query sequence(s) to the nucleotide (NT) databases in NCBI. [Learn more](#)

BLASTN contigs helps you:

- Confirm hits.
- Determine region of coverage.
- Collect metrics for publication.

Make a selection

Up to 3 of the longest contigs are available to BLAST. The more contigs you choose, the longer the results will take to load.

☐ Contig name	Number of reads	Contig length
☐ NODE_3_length_2853_cov_9.641530	465	2901
☐ NODE_26_length_371_cov_14.582278	88	378
☐ NODE_13_length_816_cov_3.897503	68	830

ContinueCancel

Genome Assembly?

- CZ ID's mNGS Nanopore pipeline is meant for metagenomic analysis only (i.e., taxonomic identification)
- The mNGS pipeline can be used to gather information for downstream genome assembly:
 - Quickly identify organisms and/or potential reference sequences
 - Download reads/contigs associated with organism of interest (FASTA)
 - Download quality- and host-filtered reads (FASTQ or FASTA)

Let's do our own analysis!

Exploring Results

- QC metrics
- Sample Report
 - Metrics
 - Analysis
- Accounting for background

Analysis Workflow - ONT

Complete a first pass sample report view

1. Look at the list of detected taxa
2. Record NT bPM, NT r, and any other relevant metrics
3. Look at the coverage plot for each taxa and record your interpretation (low, medium, high coverage)
4. BLASTn contigs from each candidate taxa and record what you find (specific, non-specific, etc)

Check negative water controls

1. Navigate to the sample report for each negative water control included in the run
2. For each taxa observed in the samples, record the NT bPM, NT r, and any other metrics in each water control
3. If you have >1 water control, calculate the average signal for each taxa

Put it all together (background subtraction + interpretation)

1. Subtract the background/water control signal from the sample signal for each taxa
2. Interpret: real signal should be higher in samples than in water controls

Go to worksheet:
Sample Report
section

Current Worksheet: CZ ID Training Reference Tables & Worksheets_ONT, pg 10-16

Next Slides: DL7-O_Background models for ONT